

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	K Ragavi	Reg. No.:	192524364
Section / Batch:	B. Tech AI&DS	Date:	22/07/2026

Code of Conduct

I K. Ragavi (192524364) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Ragavi

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Set-1

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q3 Stack 5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 int stack[5];  
3 int top=-1;  
4 int main()  
5 {  
6     printf("stack initialized");  
7     return 0;  
8 }
```

Test Input
stdin input (optional)

Output

29°C Partly cloudy

Search

Windows Taskbar

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int main()  
4 {  
5     char stack[10] = {'(', '+', '*', ')'};  
6     int top = 2;  
7  
8     while(stack[top] != '(')  
9     {  
10        printf("%c", stack[top--]);  
11    }  
12    top--;  
13  
14    return 0;  
15 }
```

Test Input
stdin input (optional)

Output

29°C Partly cloudy

Search

Windows Taskbar

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Q1

Binary Search

5 marks

✓

Q2

Stack

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Stack

5 marks

✓

Q6

Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List Data Structures

5 marks

✓

what is the error in the below code?

new.next = head
head = new

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *new = malloc(sizeof(struct Node));
12
13     new->data = 10;
14     new->next = head;
15     head = new;
16
17     printf("%d", head->data);
18
19     return 0;
20 }
```

Test Input

stdin input (optional)

Output

10

29°C
Partly cloudy

Search

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Q1

Binary Search

5 marks

✓

Q2

Stack

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Stack

5 marks

✓

Q6

Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List Data Structures

5 marks

✓

What is missing in the below pseudocode?

DEQUEUE(Q):
if front == -1:
 print("Underflow")
else:
 return Q[front++]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 int main() {
3     int q[5]={10,20,30};
4     int front = 0, rear = 2;
5     if (front == -1||front > rear)
6         printf("underflow");
7     else
8         printf("delete:%d",q[front++]);
9     return 0;
10 }
```

Test Input

stdin input (optional)

Output

delete:10

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Search

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Set-2

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Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q8 Stack 5 marks

Q9 Circular Queue 5 marks

Q10 Stack 5 marks

1/10 answered

```
1
2 #include <stdio.h>
3 #include <ctype.h>
4
5 int main()
6 {
7     char exp[50];
8     int stack[50], top = -1;
9     int i, a, b;
10
11     printf("Enter postfix expression: ");
12     scanf("%49s", exp);
13
14     for(i = 0; exp[i] != '\0'; i++)
15     {
16         if(isdigit(exp[i]))
17         {
18             stack[++top] = exp[i] - '0';
19         }
20         else
21         {
22             b = stack[top--];
23             a = stack[top--];
24
25             if(exp[i] == '+')
26                 stack[++top] = a + b;
27             else if(exp[i] == '-')
28                 stack[++top] = a - b;
29             else if(exp[i] == '*')
```

23+

Output

Enter postfix expression: Result = 5

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Search

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Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q8 Stack 5 marks

Q9 Circular Queue 5 marks

Q10 Stack 5 marks

2/10 answered

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 int dequeue()
9 {
10     int data;
11
12     if (front == -1)
13         return -1;
14
15     data = queue[front];
16
17     if (front == rear)
18     {
19         front = -1;
20         rear = -1;
21     }
22     else
23     {
24         front = (front + 1) % MAX;
25     }
26
27     return data;
28 }
```

stdin input (optional)

Output

Deleted: 10

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Search

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Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

5/10 answered

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int deque[MAX];
6 int front = -1, rear = -1;
7
8 void insertFront(int data)
9 {
10     if (front == -1)
11     {
12         front = rear = 0;
13     }
14     else if (front == 0)
15     {
16         front = MAX - 1;
17     }
18     else
19     {
20         front = front - 1;
21     }
22
23     deque[front] = data;
24 }
25
26 int main()
27 {
```

Test Input
10

Output
Enter element: Inserted element: 10

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Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8

Stack

5 marks

Q9

Circular Queue

5 marks

Q10

Stack

5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int rear = -1;
7
8 void enqueue(int data)
9 {
10     if (rear == MAX - 1)
11         return;
12
13     rear++;
14     queue[rear] = data;
15 }
16
17 int main()
18 {
19     int data;
20
21     printf("Enter element: ");
22     scanf("%d", &data);
23
24     enqueue(data);
25
26     printf("Inserted element: %d", queue[rear]);
27 }
```

Test Input

10

Output

Enter element: Inserted element: 10

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Search

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Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8

Stack

5 marks

Q9

Circular Queue

5 marks

Q10

Stack

5 marks

Your Fixed Code

```
1
2 #include <stdio.h>
3 #include <stdlib.h>
4
5 struct Node
6 {
7     int data;
8     struct Node *prev;
9     struct Node *next;
10 };
11
12 int main()
13 {
14     struct Node *head, *del, *temp;
15
16     head = (struct Node *)malloc(sizeof(struct Node));
17     head->data = 10;
18     head->prev = NULL;
19     head->next = NULL;
20
21     temp = (struct Node *)malloc(sizeof(struct Node));
22     temp->data = 20;
23     temp->prev = head;
24     temp->next = NULL;
25     head->next = temp;
26
27     del = temp;
28 }
```

Test Input

stdin input (optional)

Output

After deletion: 10

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Search

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Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8

Stack

5 marks

Q9

Circular Queue

5 marks

Q10

Stack

5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int stack[5], top = -1;
6     int i, data;
7
8     printf("Enter 5 elements:\n");
9
10    for(i = 0; i < 5; i++)
11    {
12        scanf("%d", &data);
13        stack[++top] = data;
14    }
15
16    if(top == -1)
17    {
18        printf("Underflow");
19    }
20    else
21    {
22        printf("Popped element: %d", stack[top]);
23        top--;
24    }
25
26    return 0;
27 }
```

Test Input

10 20 30 40 50

Output

Enter 5 elements:
Popped element: 50

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Search

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Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8

Stack

5 marks

Q9

Circular Queue

5 marks

Q10

Stack

5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int main()
6 {
7     int stack[MAX], top = -1;
8     int data;
9
10    printf("Enter element: ");
11    scanf("%d", &data);
12
13    if (top == MAX - 1)
14    {
15        printf("Overflow");
16    }
17    else
18    {
19        stack[++top] = data;
20        printf("Inserted element: %d", data);
21    }
22
23    return 0;
24 }
```

Run Save Submit Fix

Test Input

10

Output

Enter element: Inserted element: 10

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Search

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Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8

Stack

5 marks

Q9

Circular Queue

5 marks

Q10

Stack

5 marks

10/10 answered

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int main()
6 {
7     int queue[MAX];
8     int front = 0, rear = -1;
9     int vehicle;
10
11     printf("Enter vehicle number: ");
12     scanf("%d", &vehicle);
13
14     if ((rear + 1) % MAX == front)
15     {
16         printf("Queue Full");
17     }
18     else
19     {
20         rear = (rear + 1) % MAX;
21         queue[rear] = vehicle;
22         printf("Vehicle added: %d", vehicle);
23     }
24
25     return 0;
26 }
```

Test Input

10

Output

Enter vehicle number: Queue Full

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Search

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Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8

Stack

5 marks

Q9

Circular Queue

5 marks

Q10

Stack

5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int stack[5], top = -1;
6     int i;
7
8     printf("Enter 5 elements: ");
9
10    for(i = 0; i < 5; i++)
11    {
12        scanf("%d", &stack[++top]);
13    }
14
15    if(top == -1)
16        printf("Stack is empty");
17    else
18        printf("Top element: %d", stack[top]);
19
20    return 0;
21 }
```

Run

Save

Submit Fix

Test Input

10 20 30 40 50

Output

Enter 5 elements: Top element: 50

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Search

ENG IN

10:15 10-08-2026

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____